LLMs Can’t Keep Up with Language That’s on Fleek
How Training Data Age Stunts Model Vocabulary

Anonymous ACL submission

Abstract

The accuracy of large language models is an integral aspect of their performance. Accuracy is commonly thought of in terms of grammer and informational correctness, but also importantly encompasses the correct usage of vocabulary by models. We argue that this "correct usage" encompasses not just avoiding the use of definitionally incorrect words and phrases, but avoiding the use of irrelevant and dated vocabulary. Large language models inherit not only the knowledge but the temporal vocabulary characteristics of their training data. Because web-scale corpora carry no uniform date weighting, a model trained on a decade of internet text may treat decade-old slang as equivalent to current usage. We ask whether LLMs track the temporal lifecycle of informal vocabulary or whether their slang usage is stunted relative to the pace of real-world linguistic change. Combining a frequency analysis of slang terms in FineWeb (Penedo et al., 2024)—a filtered derivative of Common Crawl—with three complementary analyses of four LLMs, we find that (i) model slang usage is anchored to older, well-attested terms irrespective of recency. The tested models use slang with average peak usage over ten years ago; (ii) even with explicit instruction models display inaccurate awareness of peak slang usage; and (iii) models’ parametric knowledge of when a term was popular is systematically dated much earlier than it’s peak popularity based on the FineWeb corpus. We argue these effects compound as human-authored text is increasingly displaced by model output, and discuss implications for persona simulation and future training-data curation, as well as potential societal impacts from common usage of models that tend to stagnate language.

1 Introduction

1.1 Motivation: Training Data Growth and Its Limits

Large language models have scaled rapidly on ever-growing internet text corpora, and data scale remains a primary lever for capability: compute-optimal training scales tokens in proportion to parameters (Hoffmann et al., 2022), (Zhao et al., 2023). Historically the volume of available training data has grown year over year, making recency a secondary concern (Villalobos et al., 2024), (Zhao et al., 2023) **TODO:** Seems like this trend has more nuance. Two trends are shifting this picture. First, LLM-generated text constitutes a growing share of the web, threatening to contaminate future training pools with model output rather than authentic human language; Shumailov et al. (2023) show that training on model-generated content induces irreversible *model collapse*, in which the tails of the original distribution vanish. Second, if the supply of genuinely human-authored data decelerates, the median age of training data will rise until eventually the bulk of available text may be years or decades old. Both trends point to a fundemental flaw in model design: models inherently anchor their vocabulary to an earlier linguistic era and fail to track how language actually evolves.

1.2 The Temporal Language Problem

Language changes continuously, with slang and informal vocabulary changing at the fastest rate, often rising and falling within months to a few years. Because internet training corpora carry no uniform date weighting, a model trained on a decade of web text may treat decade-old slang as interchangeable with contemporary usage, due to inherent lack of awareness of the structure of their own training data. The problem is most acute for slang of the ∼2010–2020 era: it is abundant in training data yet not clearly marked as dated, so its representation is governed by total corpus frequency rather than current relevance (Section 4.1). This is compounded by misalignment in the underlying corpora: Cheng et al. (2024) show that effective training cutoffs often differ substantially from reported ones, and that Common Crawl dumps exhibit non-trivial temporal leakage, with old data persisting in newer dumps and being de facto upsampled.

In this paper we find that **LLM slang usage is temporally flat**: models do not differentiate by era when producing informal language, leaving their freeform output sounding dated due to higher representation of older slang in training corpera. We expect this effect to worsen if the bulk of high-quality, human-authored pretraining data continues to age without intervention.

1.3 Linguistic Framing

To demonstrate a temporal linguistic effect we require a set of words whose popularity has measurably risen and fallen within the span of LLM training data. We focus on *slang*, which we situate against two adjacent classes of word or phrase defined lexical change:

1. **Neologisms:** words that rise in use and plateau as they enter the standard lexicon. Our hypothesis predicts that models should also lag in adopting neologisms, but their slower, monotonic trajectories are harder to resolve on the time scale of available data. **TODO:** Add a citation defining neologisms.

2. **"Meme words" / ephemeral slang:** words that spike quickly and fade within months to a few years. Due to the current max age (3 decades) of most colloquial text training data (internet text), this time range is ideal for our study. Slang use is both creative and ephemeral (Eble, 1989; Wu and Sun, 2025), for our study we select words for their ephemerality. By selecting words that peaked in 2010–2020 and declined before our models' reported cutoffs, we can test whether reduced recent representation in training data translates into reduced usage by models.

3. **"Vogue words":** standard-English words that undergo a transient usage spike (e.g. *awesome*, *iconic*, *epic*); temporally they behave much like neologisms. This category is also examined in this study, but should be differentiated from words that increase in usage frequency due to a new definition **TODO:** (Cite Hamilton polysemy vs frequency) **TODO:** Add a citation defining vogue words.

In linguistics, slang is defined by register rather than novelty, but the definition is famously contested (Dumas and Lighter, 1978; Kipfer and Chapman, 2010). The property we rely on to define slang here is *ephemerality*: throughout this paper, "slang" denotes a word with a demonstrably ephemeral usage lifecycle in the sense of Eble (1989). Additionally, in order to demonstate the effects of models using aging slang, we cite examples of slang tied to in-group identity (Eble, 1989; Earl, 1972), in particular in-group to a specific generation (we often contrast "Millenial" and "Gen-Z" slang usage). The authors would also like to acknowledge that a substantial portion of American internet slang is appropriated African American Vernacular English or language appropriated from the Queer community; we return to this in our discussion of limitations (Section 7).

1.4 Contributions

This paper makes the following contributions:

1. A sense-disambiguated, time-resolved frequency analysis of nineteen slang terms in FineWeb, establishing a ground truth for slang prevalence in LLM training data and a proxy for the popularity of the terms at given time slices (Experiment 1).

2. A direct year-association probe showing that models' knowledge of slang era is systematically shifted into the past, dissociating what models *know* about slang age from how they *use* it (Experiment 2).

3. A semantic-slot probe showing that models reliably produce older, well-attested slang but fail on the most recent terms, and that conditioning the prompt on an explicit year does not recover era-appropriate vocabulary (Experiment 3).

4. An analysis of slang words generated by model persona simulation, mapping them back to corpus age to demonstrate a model "vintage" based on average age of slang used in a naturalistic setting (Experiment 4).

2 Background & Related Work

2.1 Training Data Sources and Temporal Coverage

Common Crawl is a major component of most large-scale training corpora and a rich source of informal language: it includes web text from platforms such as Reddit[1], whose posts and comments capture a large variety of informal language and internet slang, in contrast to sources like BookCorpus that are dominated by edited formal prose. Owing to Common Crawl's size and noise, a number of cleaned derivatives are distributed on HuggingFace using aggressive deduplication and quality filtering. We examine on one such derivative, FineWeb (Penedo et al., 2024), which provides high-quality internet text using a set of novel and established filtering methods (See https://huggingface.co/datasets/HuggingFaceFW/fineweb#data-processing-steps).

Common Crawl dumps date back to 2013 and have been collected every few months since (Common Crawl Foundation, 2007). FineWeb retains a `dump` field linking each document to its originating crawl. The repository releases pre-sampled subsets at several scales, and our analysis uses the 10-billion-token (10BT) sample as a representative slice. This sample was generated in 2024 and so spans dumps up to `CC-MAIN-2024-10` (the tenth week of 2024). The `dump` field lets us bin documents by date, typically to the granularity of a few months, and measure how the frequency of each target slang term varies over time. This is used as a corpus-level baseline for slang prevalence on the web, and a means of verifying that each target word is genuinely ephemeral (frequent in one date range and scarce in a later one). Additionally, this analysis gives us insights into which words have recently become popular and which have recently fallen out of favor.

Related Work on Emergent Behavior in Large Models Larger models show emergent capabilities that smaller ones lack (Wei et al., 2022). This is why we probe recent, capable models rather than small ones alone: small-model results may not carry over, and whatever awareness of slang recency a model has is both more likely to exist and more likely to surface in its output at scale.

Related Work on Knowledge Cutoffs and Temporal Bias A growing body of work investigates how training data temporality affects LLM behavior.

Cheng et al. (2024) find that effective cutoffs often differ from reported cutoffs due to temporal biases of CommonCrawl data and deduplication complications.

[1]Reddit closed it's API to crawlers in 2024, which is the end date of the FineWeb CommonCrawl basis in this study.

Zhu et al. (2024) find that models' understanding of world events effectively lingers around 2015 even for models with nominally later cutoffs, which is highly relevant to our hypothesis of temporal stagnance in this paper.

These works treat the knowledge cutoff as the main driver of temporal bias; We instead point to a distinct one—how much data survives from each slice of time—as a cause of model outdatedness. Additionally, our paper emphasizes the socio-cultural impact of this stagnation on slang language, as well as the implications of those effects on the growing area of persona simulation (Shao et al., 2023). **TODO:** Should add more throughout about persona simulation if I want to make this point part of the paper.

Related Work on Slang Processing and Understanding in LLMs Both Sun et al. (2024) and Wu and Sun (2025) are recent papers exploring LLM slang usage. Neither of these works examines the *temporal distribution* of slang usage, whether models favor older vs. newer slang, which is the central contribution of this paper. LLMs weaknesses in slang understanding and inability to generate slang in a way similar to human's, as demonstrated by the above papers, does provide additional support to our arguments that slang erasure is a real technical problem with models with real societal impact.

Model Collapse and LLM-Generated Data The risk of LLM-generated text contaminating future training corpora is increasingly well-documented. Shumailov et al. (2023) show that training on model-generated content causes irreversible defects where tails of the original distribution disappear.

Drayson et al. (2025) investigate model collapse under realistic conditions where data origin is unknown, and propose importance sampling as a mitigation.

Feng et al. (2024) argue that verification of synthesized data can prevent model collapse.

This body of work motivates our central concern: if model outputs are temporally frozen and increasingly populate the web, future corpora will contain an ever-growing proportion of linguistically dated text. Even where collapse is averted—e.g. by verifying synthesized data (Feng et al., 2024)—the supply of fresh human-authored text may shrink, leaving older data to dominate the high-quality and modern fraction of training corpora.

3 Methodology

Dataset: FineWeb Our corpus baseline on slang trends is drawn from **FineWeb** (Penedo et al., 2024), a large, quality-filtered, deduplicated text dataset derived from Common Crawl and widely used to train open LLMs. We use the public 10BT sample on HuggingFace, whose documents retain a `dump` field identifying the originating CC-MAIN crawl (year-week). Dumps are available from 2013 to 2024, giving us a wide range to test for word ephemerality taking place on the scale of years or months.

Working from FineWeb rather than raw Common Crawl files has two advantages. First, it already applies the quality and filtering of modern training corpora, so our frequency estimates better reflect the text models actually train on than a raw Common Crawl would. Second, the pre-extracted sample is much cheaper to process than re-crawling and parsing the full web archives. Regarding the use of FineWeb as a proxy for internet usage trends overall, while FineWeb's own filtering and deduplication may distort raw frequencies, CommonCrawl's imperfect de-deduplication processes mean that it is not a perfect representation of real internet language usage either. We therefore treat the resulting rates as a representative proxy for slang prevalence, with high relevance to LLM training, in training data rather than an exact measurement of web usage.

Identifying Target Slang Words The main analysis is divided into two main methodologies, based on the source of the set of slang words in an experiment. The initial experiments involve identifying a reference set of slang words in the FineWeb corpus, and subsequently running analysis on their frequencies across FineWeb dumps to profile their temporal characteristics. We then examine how models use and view this set of corpus identified slang words. The second set of experiments starts by prompting a set of models with persona-style prompts intended to generate free-form slang usage. We use the responses to generate a list of slang commonly used by models, which we can then run analysis against in the FineWeb corpus. This gives us chracteristics of the slang words most commonly used by the set of models, such as peak usage age in the corpus for a given word. For the set of slang words used for initial analysis in FineWeb, we draw 125 candidates from Word-of-the-Year lists, Urban Dictionary, and Google Search Trends[2]. Each candidate is then **validated against the FineWeb corpus-frequency analysis of Section 4.1** using the quantitative *ephemerality* criterion below; for the probing subset we additionally require that the word fill a recurring informal semantic slot (e.g. "impressive/cool"), keeping meaning roughly controlled across eras.

3.1 Quantitative ephemerality criterion.

We read "ephemeral" directly off the corpus trajectory. For each candidate w we take its sense-filtered rate pooled by calendar year, $r_w(y)$ for $y \in [2013, 2024]$ (Section 3.2), and compute three quantities: a *reliability* count N_w (total slang-sense hits); a *trend* given by the slope β_w and fit R^2_w of an ordinary-least-squares regression of $\log_2 r_w(y)$ on year, so β_w is the word's multiplicative drift in bits per year; and an *amplitude* $A_w = \log_2\left(r_w^{\text{peak}}/r_w^{\text{tail}}\right)$, the peak-year rate over the rate at the endpoint on the far side of the peak. A word is admitted as an ephemeral target when (i) it is well enough attested to trust its curve — $N_w \geq 100$, or $N_w \geq 50$ with an unmistakable trend ($R^2_w \geq 0.8$, $|\beta_w| \geq 0.3$) — *and* (ii) its trajectory is non-flat, i.e. at least one of: a steady directional drift ($|\beta_w| \geq 0.1$ bits/yr at $R^2_w \geq 0.5$, a $\geq 2\times$ change per decade), a prominent arc ($A_w \geq 1$ bit, a peak at least double the opposite tail), or a shallow-but-clean monotone trend ($R^2_w \geq 0.75$ with $|\beta_w| \geq 0.04$) that catches high-volume words drifting slowly. The sign of β_w together with the peak position then sorts each word into one of three classes: *declining* (early peak, $\beta_w < 0$), *rising* (late peak, $\beta_w > 0$), or *rise–fall* (interior peak, prominent on both sides).

Applied to the 125 candidates, this yields the ephemeral target set. The nineteen words carried into the probing experiments (Experiments 2–4) are drawn from it and span the three classes, mirroring the corpus-frequency figures (Figures 1–3):

1. **Pre-2018 (peaked early, now declining):** *lol, sick, troll, bro, swag, omg, meh, lmao*.
2. **Around 2020 (rising through the late 2010s):** *vibe, vibes, alpha, red pill, legit*.
3. **2022–2024 (most recent, sharply rising):** *slay, gaslight, glow-up, aura, lowkey, situationship*.

[2] Words of the Year tend not to be slang per se, but instead often relate to global events. Using Urban Dictionary and cross referencing word popularity using Google Search Trends was the most reliable way to find good candidate words.

Seventeen of the nineteen clear the criterion outright; *alpha* and *aura* are the two marginal inclusions, with shallow, noisy corpus trajectories ($R^2 < 0.45$). We keep them for semantic-slot and recency coverage — their slang senses are demonstrably recent even where the 10BT sample's thin 2024 tail understates them — but flag them as the weakest cases on corpus evidence alone. The full pass/fail table with per-word statistics is given in Appendix B.

Several of these words are polysemous (e.g. *sick*, *troll*, *aura*), with a slang sense distinct from a standard-English sense; Section 4.1 describes the sense-disambiguation step used to count only the slang usage.

3.2 Experiment 1: Corpus Frequency Analysis (FineWeb)

Goal. Establish the actual temporal frequency distribution of target slang words across time-sliced web data, serving as a ground truth against which LLM output distributions can be compared.

Data source. Rather than analyzing raw Common Crawl, we use the **FineWeb** (Penedo et al., 2024) 10BT sample hosted on HuggingFace, which carries a `dump` field tagging each document with its originating CC-MAIN crawl (year-week), allowing us to slice the sample by crawl date. FineWeb is itself a filtered, deduplicated derivative of Common Crawl, so it both reflects the kind of corpus used to train modern LLMs and removes much of the boilerplate and duplication that makes raw Common Crawl analysis difficult and unreliable. FineWeb processing provides data spanning 2013–2024, which matches with the published cutoff date for our open models used.

Sense disambiguation. A central difficulty is that many target words are polysemous: *sick* (ill vs. excellent), *troll* (creature vs. online provocateur), *aura* (energy field vs. Gen-Z coolness). To address disambiguating the slang and non-slang senses during counting, we extract the textual context of every target-word occurrence and score it with a **RoBERTa classifier fine-tuned to distinguish the slang sense** from the standard sense. Only occurrences scoring above a threshold (≥ 0.99, selected to minimize false positives) are counted as slang usage.

Normalization. Sense-filtered counts are converted to a rate expressed as occurrences per million tokens of the corresponding dump:

$$r_{w,d} = \frac{\text{count}_{\text{slang}}(w,\, d)}{\text{tokens}(d)} \times 10^6$$

where *w* is a target word and *d* is a crawl dump. Per-dump token totals are taken from the FineWeb sample metadata. This normalization controls for the fact that successive dumps vary substantially in size.

Uncertainty. The 10BT sample is a random draw from FineWeb, so a per-dump occurrence count k is modeled as $k \sim \text{Poisson}(r_{w,d} \cdot \text{tokens}(d))$. We report the exact Poisson interval (95% by default) as a shaded band around each word's rate curve; bands are wide for rare words and tight for common ones.

Output. For each target word we plot $r_{w,d}$ against dump date on a $\log_2$ scale, producing the temporal frequency curves in Section 4.1. These curves are the corpus ground truth against which the LLM probing distributions are compared.

TODO: Document the RoBERTa sense-classifier training data and held-out accuracy (`FineWebAnalysis/finetune_roberta.py`). Report the FineWeb sample token totals per dump used as denominators.

3.3 Experiment 2: Direct Year Association

Goal: Experiment 3 measures *usage* — what slang the model reaches for when it is not told to think about time. Experiment 2 instead measures *meta-knowledge*: when asked directly, does the model know *when* a slang word was popular? Contrasting the two isolates whether failures are a knowledge gap or a usage bias. This is the inverse of the Experiment 3 design constraint: here we deliberately provide a temporal cue via the prompt.

We use three prompt forms, each rendered once per target word and sampled across all four models:

1. **GiveYear:** *"The slang word '{word}' is most associated with what year? Respond using only the year."*: directly requesting the word's peak year.

2. **DatePhrase:** *"You hear someone saying '{word}' in conversation, which year are they most likely saying this in?"*: a softer, scenario-framed variant of the same question. **TODO:** I think we may want to cut this experiment. logprob feels more robust, but overal seems redundant.

GiveYear and DatePhrase are sampled five times per word per model, we then parse a four-digit year from each free-text response. Comparing the model's stated peak year against the corpus peak from Experiment 1 tells us whether a model's *declarative* knowledge of slang timing is accurate.

3.4 Experiment 3: Semantic Slot Querying

Goal: Measure what slang a model actually *produces* when asked for an informal word, with no temporal cue in the prompt, so we observe natural usage rather than meta-knowledge about eras. A model whose vocabulary tracked current usage should readily produce recent slang; a temporally frozen model should fall back on older, well-attested terms. A central design constraint is that we avoid naming years or providing informal language queues in the prompt, which would test meta-knowledge or register-matching instead of general usage.

We give the model a temporally neutral semantic description of a meaning and ask for a single informal word. The probe runs in two modes:

Concept-slot mode. The description names a general meaning with several era-spanning valid answers ("something impressive/cool", "mediocre", "charisma", "suspicious", . . .) and we record which word the model reaches for, surfacing its default vocabulary per slot without naming any target word.

Target-recovery mode. The description is written to point at one specific target word from Section 3.2 — e.g. a definition that uniquely evokes *gaslight* or *aura* — and we measure how often the model returns that exact word, counting stem and inflection matches (so *glow-up*/*glowup* and *gaslight*/*gaslighting* all count). Recovery rate is a direct measure of how reluctant the model is to produce the intended term, and of whether the term is effectively out of vocabulary. Each target word is probed with three independently worded phrasings to control for wording, and each phrasing is sampled ten times per model.

TODO: This experiment should probably move to the Experiment 4 section.

Year-conditioned variant. To test whether a model conditions its vocabulary on a stated time, we prepend "The year is {year}." to each target-recovery prompt and sweep {year} over 2013–2025. If a model were aware of its own data distribution, we would expect recovery accuracy for a word should be highest at, or near, the year of that word's true corpus peak (Experiment 1). Flat accuracy across years would indicate the model ignores the temporal cue; a peak at the wrong year would indicate a miscalibrated sense of timing, or one not based on true distribution.

We probe four models spanning three providers: two open-weight models served via Together AI, `meta-llama/Llama-3.3-70B-Instruct-Turbo` and `deepseek-ai/DeepSeek-V4-Pro`, and two closed models, `gpt-4o` (OpenAI) and `claude-sonnet-4-6` (Anthropic). Because the closed APIs do not expose token logprobs, responses are sampled (temperature=1.0) and summarized over samples by frequency. Notably, the Llama model includes RLHF, which as we see in the results may suppress slang and act as a confounding factor.

Experiment 4: Persona Prompting **Goal:** The other queries all pin the model down to a single-word answer. Experiment 4 attempts to invert the goal of querying information about known slang words from the models, instead seeking to find what slang is commonly generated by models and then analyzing the characteristics of that set of words. Additionally, with the increased popularity of persona work using models **TODO:** citations, this method provides an analysis more directly relevant to current LLM applications.

Prompts. We give the model a persona instead of a question. The system prompt tells it to act like "a young person using informal language like you would among peers," and two user prompts ask it to simulate a short text-message or group-chat exchange between friends. The prompt never names a target word, a year, or even the word "slang," so whatever informal vocabulary shows up is volunteered rather than elicited. We collect 100 free-form responses per model (400 in total) at temperature 1.0.

Scoring via effective vintage. To turn free text into a temporal measurement, we scan each response for words from a 150-item slang lexicon (generated by combining the words from section 1 with manually extracted phrases/variants from this experiment's responeses) and tag every hit with that word's corpus peak year [3]. We then summarise a model's output by its *effective vintage*: the

[3] Peak year is calculated using the word counts generated in section 1. For words added to the set from the experiment 4 responses we re-ran an identical set of steps to initial slang collection from section 1 to find words counts per dump.

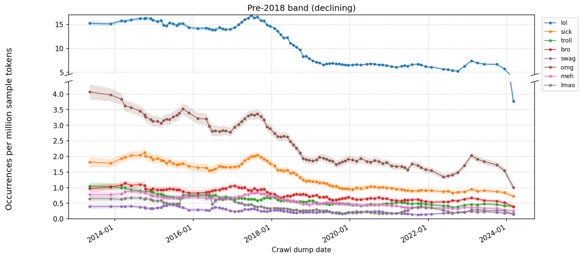

Figure 1: Slang-sense occurrences per million tokens by FineWeb crawl dump (linear scale, RoBERTa score ≥ 0.99, 5-dump smoothing, 95% Poisson CI) for the pre-2018 declining band (*lol, sick, troll, bro, swag* plus the additional terms *omg, meh, lmao*). The y-axis is broken so *lol* (peak ∼16/M) sits in its own upper panel and does not flatten the rest: *lol*, *omg*, and *sick* decline steadily after ∼2018 and *swag* drifts downward, while *troll*, *bro*, *meh*, and *lmao* sit lower and taper more gently.

occurrence-weighted mean peak year of the slang it produced. A model's datedness can be measured by how far behind present day it's *effective vintage* lies. As a baseline we compute the same weighted mean over the lexicon's corpus frequencies (its *corpus vintage*), so the reference point is the corpus's own centre of mass rather than the calendar. This frame of reference serves to confirm that the model does not exhibit emergent behavior that might bring it's slang usage closer to current trends. Appendix H gives the estimator, its bootstrap confidence intervals, and a distributional cross-check.

4 Analysis

4.1 Experiment 1: Corpus Frequency Over Time

Slicing the FineWeb 10BT sample by crawl dump and applying the RoBERTa sense filter yields, for each target word, a curve of slang-sense occurrences per million tokens against dump date (2013–2024) with 95% Poisson confidence bands. These curves confirm that the target words exhibit the ephemeral lifecycles the probing experiments rely on, and they separate cleanly into the three temporal bands of Section 3.2, which we present in turn (Figures 1–3).

Two properties matter for the following probing analysis. First, the words we treat as "older" (*lol*, *swag*) are demonstrably in decline by the end of the window, while the "recent" words (*aura*, *slay*, *glow-up*) are still rising. A true-to-life model of slang usage would emphasize using more recent and relevant slang words, while deemphasizing older slang that has become dated and much less popular. Sec-

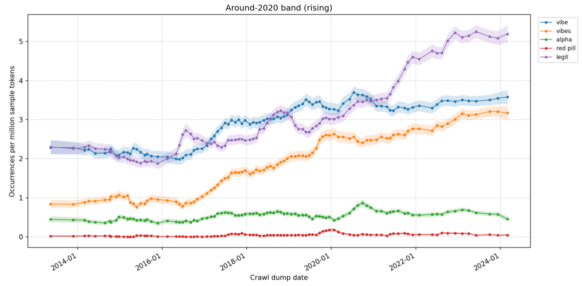

Figure 2: Corpus frequency for the around-2020 rising band (linear scale, RoBERTa score ≥ 0.99, 5-dump smoothing, 95% Poisson CI): *vibe, vibes, alpha, red pill* plus the additional term *legit*. *vibe*, *vibes*, and *legit* rise through the late 2010s and into the 2020s; *alpha* stays low and roughly flat, and *red pill*, a rarer and noisier term, sits near the floor with correspondingly wide confidence bands.

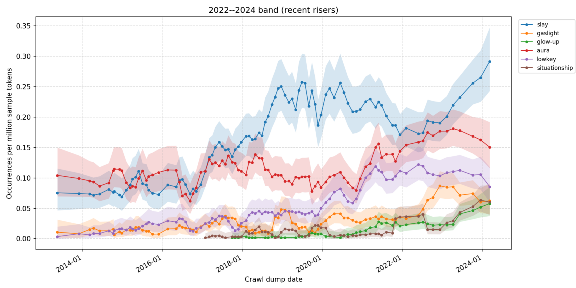

Figure 3: Corpus frequency for the 2022–2024 recent-riser band (linear scale, RoBERTa score ≥ 0.99, 5-dump smoothing, 95% Poisson CI): *slay, gaslight, glow-up, aura* plus the additional terms *lowkey, situationship*. *slay* and *lowkey* climb over the window, and *gaslight*, *glow-up*, and *situationship* appear only near its end — slang that postdates the bulk of the models' training data, though still within the models' training windows. *aura* (in its Gen-Z slang sense) is noisier and rises less cleanly.

ond, the sense filter is what makes these counts meaningful: without it, *sick* and *aura* would be dominated by their standard-English senses and show no slang lifecycle at all. Additional details regarding the sense filtering process, such as model recall and precision, are included in the appendix.

4.2 Experiment 2: Direct Year Association

Experiment 2 asks what the models *know* about slang timing, by posing the temporal question directly (Section 3.3). We see that while models carry *directionally* correct declarative knowledge of when slang was popular, that knowledge is both noticeably shifted toward the past and not reflected in their spontaneous usage (Experiment 3).

The figure overlays model belief against corpus ground truth. Figure 4 places the target words on the y-axis *ordered by their true corpus peak year* (from Experiment 1), with calendar year on the x-axis. Because the words are sorted by true peak, the stars form a rising staircase, so the horizontal offset of each diamond from its star reads directly as how far the model is off.

Models order slang roughly by era but compress it toward the past. The diamonds broadly follow the staircase — models do place *lol*, *troll*, and *bro* early and *aura* and *slay* late — but they systematically date the newer terms too early. For `claude-sonnet-4-6`, *alpha* (corpus peak 2020) is placed near 2010, and *vibe*/*vibes* (still rising through 2024) near 2018/2020. The mean absolute error against the corpus peak is sizeable for every model: 6.5 years for Claude, 7.1 for Llama, and 10.3 for DeepSeek, consistent with a sense of slang anchored to an earlier window than current usage.

Caveat on the comparison. The corpus "peak" is the year of maximum frequency, so for the words still rising at the end of the data window (2023–2024 peaks) it is closer to a lower bound on recency; a model dating *vibe* to 2018 is arguably reporting the *onset* of popular use rather than the corpus peak. The comparison is cleanest for words that have visibly crested within the window (*lol*, *swag*, *alpha*), where the models still skew early. `gpt-4o` is a special case: its lower 5.5-year error is an artifact rather than better calibration, since its *DatePhrase* responses collapse to "2023" (close to its cutoff date) for almost every word, which happens to sit near the many recent peaks.

Interpretation. The models possess *directional* temporal knowledge but that knowledge is both

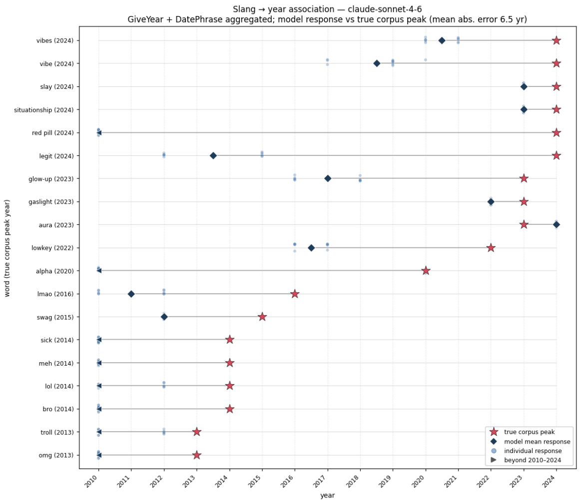

Figure 4: Direct year association for `claude-sonnet-4-6`, pooling the *GiveYear* and *DatePhrase* prompts. Words (y-axis) are ordered by their true corpus peak year (Experiment 1, shown in parentheses); the x-axis is calendar year. For each word the red star is the corpus peak, the dark diamond is the model's mean responded year, the connecting line is the error, and faint dots are individual responses; markers falling outside 2010–2024 are drawn as arrows at the axis edge. The diamonds track the rising staircase of true peaks only loosely and skew early for the newer terms (mean absolute error 6.5 years). Per-model figures for `gpt-4o` (5.5 yr, inflated by its collapse to "2023"), `DeepSeek-V4-Pro` (10.3 yr), and `Llama-3.3-70B` (7.1 yr) are included as supplementary figures (`give_year_date_phrase_*.png`).

compressed toward the past and, critically, not carried through in their actual usage (Experiment 3). The failure to produce current slang is therefore not a total absence of temporal knowledge, but shows the divergence between model's declarative knowledge and behavior, a particularly important discrepancy for evaluting how models may interact with language trends.

4.3 Experiment 3: Semantic Slot Querying

This experiment is our first framework for querying models to explore their awareness of slang's temporal characteristics. We generate a selection of prompts that are intended to solicit one of a set of target words from the models, and test their ability to predict the intended word. The list of target words is the set of slang words from section one, with known temporal characteristics.

Target-word recovery **Target-recovery accuracy varies sharply by word, not just by model.** Figure 5 shows, for each model, the recovery accuracy of every target word across the swept prompt years (ten samples per phrasing, three phrasings per word). Aggregate recovery, pooling over words and years, is highest for `claude-sonnet-4-6` (69%), followed by `DeepSeek-V4-Pro` (63%) and `gpt-4o` (59%), with `Llama-3.3-70B` lowest at 52% — consistent with RLHF-driven slang suppression in the Llama model. Unambiguous, well-established terms (*lol*, *troll*, *gaslight*, *glow-up*, *vibes*) are recovered near-perfectly by every model, whereas polysemous, subtle, or contested slots are hard across the board: *sick* (models answer the literal "awesome"/"cool" rather than the slang *sick*), *lowkey* (models return the plain paraphrase "secretly"/"kind of"), *lmao* (models fall back on *lol* or "haha"), and the newest term *aura* (models reach for *clout*, *charisma*, or *vibe*) are each recovered barely one time in ten; *omg*, *legit*, and *swag* fare only modestly better.

The hardest slots are the most diagnostic. That *aura* — the newest term, still sharply rising in the corpus (Figure 3) — is among the words models are least able to produce on demand is consistent with the central hypothesis: the slang that is most current in human usage is the slang the models are least fluent in. Conversely, the words recovered most reliably are the older, well-attested terms that have saturated the training data.

Conditioning on a stated year The year-conditioned variant prepends "The year is {year}." to each recovery prompt and sweeps the year over 2013–2025. If a model tracked its own data distribution, recovery for a word should be highest at that word's true corpus peak year. Figure 6 shows the result as a word $\times$ prompt-year accuracy heatmap for `claude-sonnet-4-6`, with rows ordered by true corpus peak and each word's true peak year outlined.

The stated year has little effect. For the majority of words (*troll*, *bro*, *lol*, *gaslight*, *vibe*, *vibes*) accuracy is essentially flat (near 100%) across all thirteen prompt years: changing "2013" to "2024" leaves the output unchanged. The model is not conditioning its vocabulary on the stated time, even when the stated year is one that has fallen out of common use. **TODO:** This is very unconvincing, the slots here are all words that fit very well, sometimes unambiguously. Would love to redo this experiment using an A/B setup to show more frequent corpus words are favored, not recent ones. Or use a list to choose from e.g. sick, dank, rad, lit, hype, etc.

Where accuracy does vary, it does not track the true peak. Among the words with year-to-year variation, the high-accuracy cells do not line up with the outlined true-peak cell. The starkest case is `gpt-4o`'s *red pill* (true corpus peak 2024): recovery is *highest* in the earliest prompt years ($\sim$90% at 2013) and *lowest* at the true peak (20% at 2024). Claude's *alpha* (peak 2020) is recovered at 100% through 2013–2017 but dips to $\sim$67% around its actual peak. If anything, models recover a word slightly better when told an *earlier* year, which matches the backward-shifted sense of timing quantified in Experiment 2.

Taken together, the two views give a consistent reading: models reliably produce older, well-attested slang but struggle with the most recent terms, and an explicit year cue does not shift them toward era-appropriate vocabulary. **TODO:** Add the concept-slot result (which default word each meaning slot evokes, coloured by era) once that figure is regenerated for the four-model panel, and quantify the year-vs-peak alignment with a per-word correlation statistic.

4.4 Experiment 4: Persona Prompting

Experiment 4 inverts the direction of investigation, starting with relatively free form "persona" generations by the models, and analysing the generated vocabularies temporal properties. Figure 7 counts how often each lexicon word appears across the

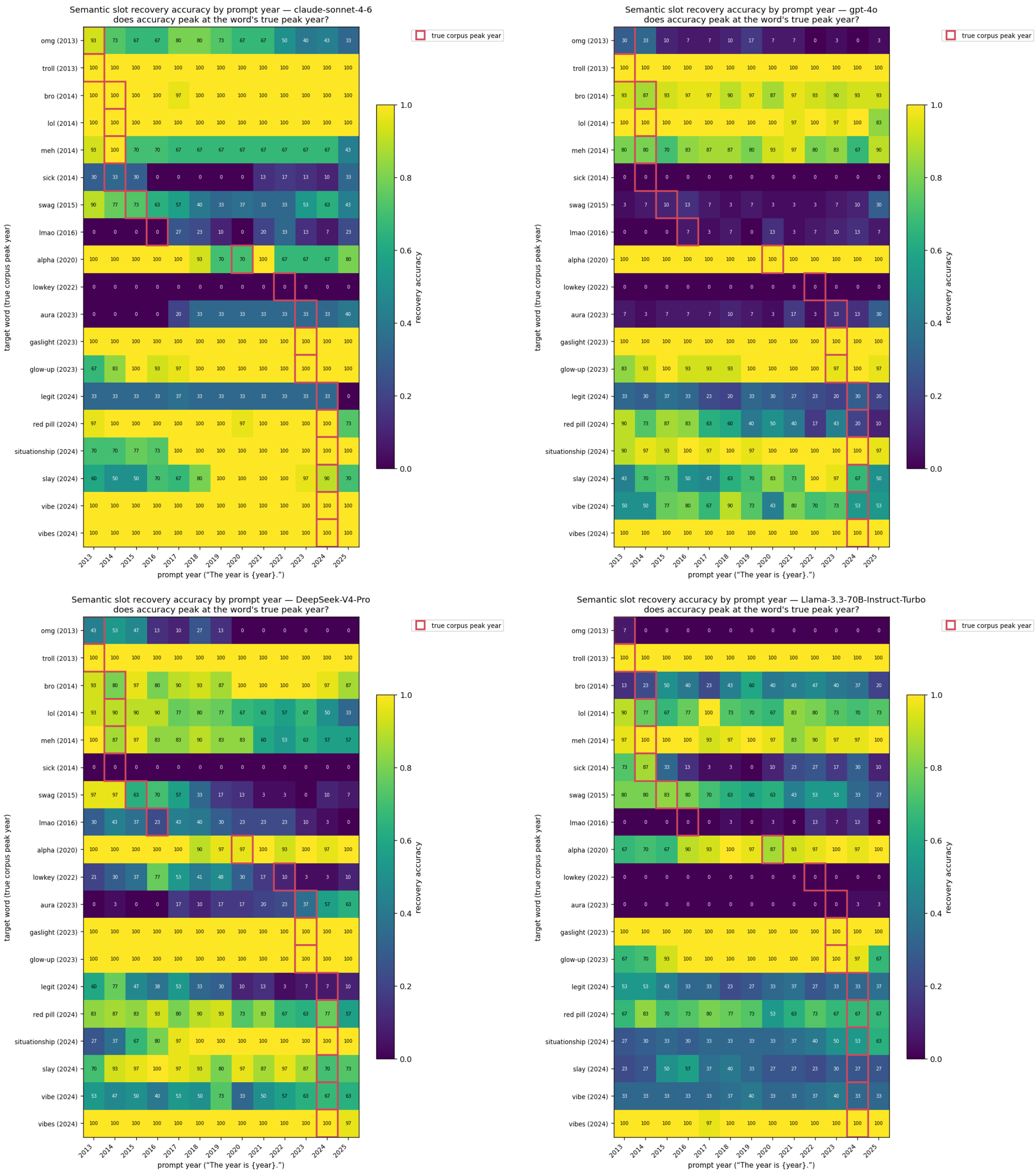

Figure 5: Semantic-slot target-word recovery accuracy for all four models: `claude-sonnet-4-6` and `gpt-4o` (top row), `DeepSeek-V4-Pro` and `Llama-3.3-70B` (bottom row). Within each panel, a cell is the recovery accuracy for a target word (y-axis, ordered by true corpus peak year, shown in parentheses) when the prompt is prefixed "The year is {year}." for the year on the x-axis (30 samples per cell; stem and inflection matches counted), and each word's true corpus peak year is outlined. Reading down a column shows which words a model can produce; reading across a row shows the (near-absent) effect of the stated year. Older, unambiguous terms are recovered reliably across all years, while polysemous (*sick*, *swag*), subtle (*lowkey*), and very recent (*aura*) terms are recovered poorly — a pattern that holds across all four models.

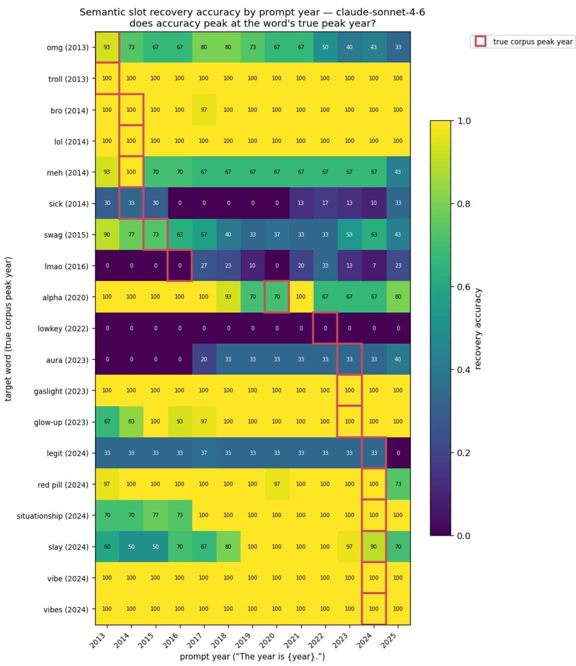

Figure 6: Year-conditioned semantic-slot recovery for `claude-sonnet-4-6`: each cell is recovery accuracy for a target word (y-axis, ordered by true corpus peak year, shown in parentheses) when the prompt is prefixed "The year is {year}." for the year on the x-axis (30 samples per cell). Each word's true corpus peak year is outlined in red. If the model conditioned on the stated year, the bright cells would track the red outlines; instead most rows are flat (the year is ignored) and the rows that do vary do not align with the true peak. Per-model figures (`target_word_accuracy_by_year_*.png`) show the same pattern, with `gpt-4o`'s *red pill* the clearest anti-alignment.

400 persona responses, ordered by corpus peak year (top panel) and by corpus frequency (bottom). A few long-lived, high-frequency terms dominate: *u*, *lol*, and *omg* together make up roughly 79% of all in-lexicon slang, and adding *bro* brings the four terms that peaked in 2013–2014 to about 84% of the total. Recent slang, such as *vibe*, *slay*, *legit*, *aura*, turn up in the low single digits or not at all, echoing the recovery gap from the semantic-slot probe.

The output carries an early-2010s vintage. Putting a number on the skew, the slang these models emit has an occurrence-weighted effective vintage of **2014.0** (95% CI $[2013.9, 2014.1]$; Poisson bootstrap), roughly **1.7 years older** than the **2015.8** vintage of the same lexicon in the corpus (Figure 8). Every model skews old, so this is not a single outlier dragging the average down. Notably, we do not observe any bias in the slang toward generating more recent slang words to better fit with trends in language. Instead we unsurprisingly see a relatively close match to the distribution in the training data, with a tendency to overuse phrases that show recent decreases in popularity (lol, omg). **TODO:** This could be good stats to include but I need to understand what exaclty its showing better. A distributional cross-check — fitting the produced-year distribution as a mixture over calendar years — agrees, piling most of the weight into the mid-2010s and cutting the divergence from a flat baseline by about a quarter (Appendix H).

Reluctance versus recency. The models differ mainly in how much slang they volunteer at all: `gpt-4o` produces the fewest in-lexicon hits of the four (205, against 881 for `claude-sonnet-4-6`), in keeping with its general reluctance to slip into a casual register unprompted (). What none of the four do is reach for current slang; the early-2010s skew holds across all of them.

5 Results

Findings from the corpus analysis and the probing experiments converge on a consistent pattern:

1. **The target words have genuine, separable lifecycles.** The corpus analysis (Experiment 1, Figures 1–3) confirms that the older terms (*lol*, *swag*) are in decline by the end of the window while the newest terms (*aura*, *slay*, *glow-up*) are still rising. This is what makes the words usable as a temporal probe and informs the curated target set selection.

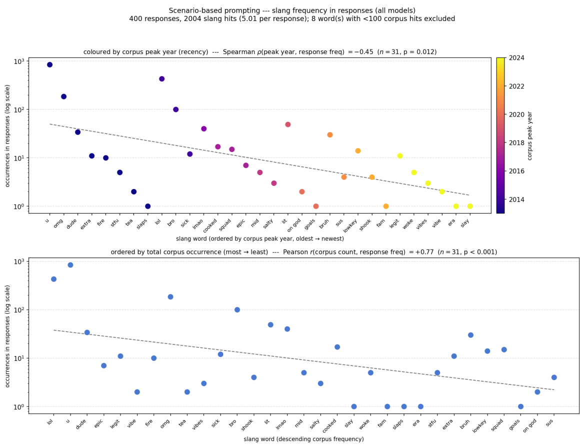

Figure 7: Slang word frequency in naturalistic persona-simulation responses (log scale), aggregated across four models (400 responses, 100 per model). Top panel: words ordered by corpus peak year (oldest → newest), coloured by peak year (plasma scale). Bottom panel: same words ordered by descending corpus frequency. Dashed lines are log-linear trend fits. High-frequency, older terms (*u*, *lol*, *omg*) dominate both panels; recent slang appears at the far right of the top panel with counts near 1, reproducing in free generation the recency gap observed in forced-choice recovery.

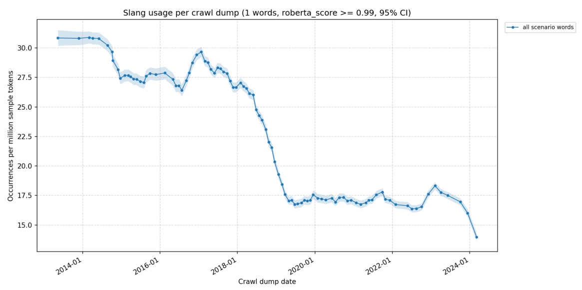

Figure 8: Corpus baseline for the persona lexicon: aggregate slang-sense frequency across the full scenario vocabulary in FineWeb 10BT, normalized per million tokens per crawl dump and summed across all words. The aggregate curve rises through the mid-2010s and plateaus, reflecting the combined lifecycle of the informal lexicon the models draw on rather than any single word's arc. This is the corpus-side reference whose centre of mass (a *corpus vintage* of 2015.8) the models' 2014.0 output vintage is measured against.

TODO: Need new recent examples potentially, seems like polysemy is counfounding.

2. **Models are least fluent in the most current slang.** In semantic-slot recovery (Experiment 3, Figure 5), every model recovers older, unambiguous terms (*lol*, *troll*, *gaslight*, *glow-up*) near-perfectly, but recovery collapses for the newest term *aura* and for polysemous slots (*sick*, *swag*) where the model prefers a competing or literal word. The slang that is most current in human usage is the slang the models are least able to produce on demand.

3. **An explicit year cue does not fix it.** Prefixing the recovery prompt with "The year is {year}." (Experiment 3, Figure 6) barely changes which slang the model produces: most words are recovered at a flat rate across 2013–2025, and the words that do vary do not peak at their true corpus-peak year — for the most recent terms (*red pill*, *vibe*) accuracy is, if anything, highest when an *earlier* year is named. The stated date does not seem to influence the models' confidence in choosing out-of-date slang terms.

4. **Parametric knowledge vs usage are unaligned.** The year-association queries (Experiment 2, Figure 4) show that models order slang by era directionally correctly, but place the newer terms years too early relative to the corpus peak (mean absolute error 5.5–10.3 years per model), and their spontaneous usage does not track recency at all. The temporal failures contain both a usage bias and a backward-shifted sense of slang timing.

5. **Model-level differences are consistent with RLHF slang suppression.** `Llama-3.3-70B` has the lowest semantic-slot recovery (52% vs. 69% for Claude) and most readily defaults to neutral standard-English words, consistent with reports that RLHF biases toward Standard American English (Barnhart et al., 2025). `gpt-4o` exhibits a distinct artifact (returning the same year for all word association queries) that is orthogonal to the era bias but worth noting. **TODO:** Confirm whether DeepSeek/Claude RLHF pipelines differ enough to explain the recovery-rate spread, or maybe fully drop this point. Bias

against colloquial language is pretty much still consistent.

6. **The skew survives in free generation.** When the models are prompted to write casually in a persona (Experiment 4, Figure 7), the slang they volunteer still skews old. Occurrence-weighted effective vintage (as outlined in Experiment 4's methodology section) is 2014.0, about 1.7 years behind the 2015.8 vintage of the same lexicon in the corpus, and a handful of long-lived terms (*u*, *lol*, *omg*, *bro*) account for the bulk of usage.

6 Discussion

6.1 Interpretation

Results suggest the models are not sensitive to temporal trends in their spontaneous slang usage. When asked to produce current slang on demand they reliably return older, well-attested terms and struggle most with the newest ones (Experiment 3). An explicit year cue does not move them toward era-appropriate vocabulary, even while they retain *directionally* correct declarative knowledge of when those terms were popular (Experiment 2). This is consistent with the hypothesis that model slang *usage* reflects total frequency in the training corpus rather than a temporally calibrated view of what is current. While this may be an intuitive result, we argue that the implications of this quality of LLMs has sever downstream implications for the model ecosystem and its societal impacts. **TODO:** Strengthen the framing with linguistics citations on the social and identity-signaling function of slang, motivating why temporally miscalibrated informal vocabulary matters independently of the downstream training-data argument.

The difficulty models have producing the most recent term *aura* (which is still sharply rising in the corpus at the end of the window in mid 2024) is striking precisely because the same models date it to roughly the right era (2023–2024) when asked directly (Figure 4). *aura* is the cleanest illustration of usage-bias-without-a-knowledge-gap: the model knows when the word belongs yet underrepesents it in generation. For most other recent terms the model additionally shifts its stated date years too early, so there is unclear declarative knowledge in addition to miscalibrated usage in generation. The reliability gap between recovering older versus newer slang may reflect that the older terms have saturated the training data with strong, redundant signal, whereas the newest terms appear in quantities too small (and too close to the data-collection horizon) to anchor confident generation, despite being the dominant form in current human usage.

6.2 Implications for Training Data

If models cannot keep pace with linguistic change, the effect compounds as web text becomes increasingly model-generated (Shumailov et al., 2023): output reflecting stagnant, older vocabulary re-enters future training corpora, further entrenching dated language and eroding models' capacity to represent current human usage. This connects to the "stochastic parrots" critique (Bender et al., 2021)—models capture the statistical structure of language but need not track its dynamics, leading to a fundementally inaccurate view of human language.

The practical effect today is modest, and is partly masked by instruction-tuned models' general reluctance to produce slang at all (Section 4.3) (which we would argue has further social implications as young people get a higher and higher percentage of their linguistic contact from models). The longer-term concern is both larger and more general than slang: that models lagging ability to reflect current usage trends makes them a perpetually outdated interface, and inaccurate representations of language (or by extension personas). As models are retrained on the accumulated text of past decades—increasingly including their own outputs—the language they produce may drift steadily further from contemporary usage. Mitigating this will require explicit training interventions, potentially things like recency-weighted sampling, together with a sustained supply of fresh, verifiably human-authored text. Our results add weight to the case for treating human-generated text as a premium training resource critical to maintaining models that use language accurately.

7 Limitations

1. Cross-model comparisons confound knowledge cutoff with architecture, training data, and alignment procedure (e.g. RLHF); we therefore emphasize within-model temporal patterns over cross-model magnitudes.

2. All four probed models are instruction-tuned. RLHF and instruction tuning are known to suppress non-standard English (Barnhart et al., 2025) and may amplify the apparent temporal freeze. Replicating on base models would

isolate this effect, but would limit us to experimenting on open models instead of more popular close ones.

3. Much American internet slang is appropriated African American Vernacular English. We do not analyze this dimension, though documented model bias against non-standard English (Barnhart et al., 2025) likely interacts with the effects we report.

4. Quality filtering and deduplication in Common Crawl derivatives can distort raw term frequencies; our FineWeb-based rates inherit any such distortion, and the de-facto upsampling of older dumps (Cheng et al., 2024) biases the corpus baseline toward earlier usage. Mitigation of this is sought by the FineWeb filtering methodologies. Additionally, the conclusions here are concerned with the pragmatic effects of the training data on model language, so distortions in training data do not detract from the conclusions.

8 Future Work

TODO: Both of these are actually things that should likely be addressed in this paper. Not really future work I think.

Fine-tuning as mitigation. The most direct test of our framing is interventional: fine-tune a model on a recency-weighted subset of the data (either upsampling recent dumps, or downsampling older ones) and re-run the queries. If the temporal freeze is driven by aggregate corpus frequency, recency weighting should shift recovery toward newer slang, whereas no change would point to a deeper architectural or alignment cause. **TODO:** Review the literature on recency-weighted and continual-learning training for LLMs before designing this experiment.

Register-controlled free generation. Experiment 4 (Section 4.4) is a first pass at measuring slang usage in free generation, but it leans on a single fixed persona and does not control for prompt register or scenario. A stronger version would systematically vary the register (formal vs. contemporary-casual) and the conversational scenario, and report per-era slang production rates against matched human baselines, to pin down how much of the old-slang skew comes from the model versus the prompt. **TODO:** Run a controlled free-generation probe and report per-era slang production rates with matched register prompts.

Further directions.

1. Sub-year temporal resolution: can models distinguish slang peaks within a single year (e.g. fast-moving meme words)?

2. Replicating the queries on base (non-instruction-tuned) checkpoints to isolate the contribution of RLHF to slang suppression.

3. A longitudinal study tracking vocabulary drift as models are retrained on successive crawl snapshots (resource-intensive, as it requires repeated pretraining).

4. Multilingual analysis: low-resource languages may be affected more strongly (model output can quickly dominate scarce human data) or less (smaller corpora may be more easily refreshed by recent text).

Acknowledgments

TODO: Fill in acknowledgments before camera-ready submission.

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A Target Words and Corpus Peaks

Table 1 lists the nineteen probed slang terms, the temporal band they belong to, and their true corpus peak year — the calendar year of maximum slang-sense frequency in the FineWeb 10BT sample (Experiment 1, RoBERTa sense score ≥ 0.99). The band names the era in which a term rose to prominence; the peak year is the corpus argmax. The 10BT sample ends at dump `2024-10`, so for terms still rising at that horizon (*vibe*, *vibes*, *red pill*, *slay*) the listed peak is the final available year and should be read as a lower bound on recency.

Word	Band	Peak	Peak rate	Hits
troll	pre-2018	2013	1.008	6,077
bro	pre-2018	2014	1.043	7,914
lol	pre-2018	2014	16.162	107,305
sick	pre-2018	2014	2.017	14,015
swag	pre-2018	2015	0.408	2,624
omg	pre-2018	2013	4.125	24,593
meh	pre-2018	2014	0.844	5,775
lmao	pre-2018	2016	0.716	3,816
alpha	around 2020	2020	0.702	5,764
vibe	around 2020	2024	3.878	30,862
vibes	around 2020	2024	4.003	20,311
red pill	around 2020	2024	0.135	546
legit	around 2020	2024	5.350	32,897
gaslight	2022–2024	2023	0.072	316
glow-up	2022–2024	2023	0.041	98
aura	2022–2024	2023	0.178	1,231
slay	2022–2024	2024	0.308	1,795
lowkey	2022–2024	2022	0.114	567
situationship	2022–2024	2024	0.077	121

Table 1: Probed target words, temporal band, corpus peak year, peak-year slang-sense rate (occurrences per million tokens), and total slang-sense hits across the FineWeb 10BT sample (RoBERTa sense score ≥ 0.99). Peak year is the argmax, over calendar years, of the token-normalised yearly rate; "Hits" is the raw count of above-threshold occurrences pooled over all 95 dumps. Rates span nearly three orders of magnitude, from the saturated *lol* (16.2 per M) to the emergent *glow-up* (0.04 per M).

B Ephemerality Criterion: Per-Word Statistics

Table 2 reports the trajectory statistics behind the ephemerality criterion of Section 3, for all nineteen probed words. For each word we regress $\log_2 r_w(y)$ on calendar year over the sense-filtered, year-pooled rates: β is the slope (bits/yr), R^2 the fit, A the peak-to-opposite-tail amplitude (bits), and N the total slang-sense hits. A word passes when it is reliably attested ($N \geq 100$, or $N \geq 50$ with $R^2 \geq 0.8$ and $|\beta| \geq 0.3$) *and* non-flat ($|\beta| \geq 0.1$ at $R^2 \geq 0.5$, or $A \geq 1$, or $R^2 \geq 0.75$ with $|\beta| \geq 0.04$). Seventeen of the nineteen pass; *alpha* and *aura* fail on a shallow, low-R^2 trajectory and are retained on semantic-slot grounds.

Word	Class	N	Peak	β	R^2	A	Pass
troll	decline	6,077	2013	−0.12	0.90	1.26	yes
bro	decline	7,914	2014	−0.13	0.79	1.24	yes
lol	decline	107,305	2014	−0.19	0.87	1.68	yes
sick	decline	14,015	2014	−0.14	0.90	1.33	yes
swag	decline	2,624	2015	−0.10	0.64	0.82	yes
alpha	rise	5,764	2020	+0.05	0.43	0.73	no
gaslight	rise	316	2023	+0.24	0.85	1.42	yes
glow-up	rise	98	2023	+0.58	0.87	5.36	yes
aura	rise	1,231	2023	+0.04	0.21	0.75	no
vibe	rise	30,862	2024	+0.08	0.81	0.75	yes
vibes	rise	20,311	2024	+0.21	0.94	2.09	yes
red pill	rise	546	2024	+0.27	0.63	2.87	yes
slay	rise	1,795	2024	+0.18	0.83	1.99	yes
Additional terms (added to the probing set)							
omg	decline	24,593	2013	−0.17	0.86	1.66	yes
meh	decline	5,775	2014	−0.16	0.88	1.60	yes
lmao	decline	3,816	2016	−0.19	0.78	1.81	yes
legit	rise	32,897	2024	+0.13	0.89	1.22	yes
lowkey	rise	567	2022	+0.35	0.80	3.83	yes
situationship	rise	121	2024	+0.60	0.88	6.27	yes

Table 2: Trajectory statistics behind the ephemerality criterion (Section 3). N: total slang-sense hits; β: $\log_2$-rate slope (bits/yr) with fit R^2; A: peak-to-opposite-tail amplitude (bits). Rows are the original thirteen probed words (top, by peak year) and the six later additions (bottom); all nineteen are probed. *alpha* and *aura* are the only words that fail, on a shallow, low-R^2 trajectory.

C Experiment 1: FineWeb Corpus Frequency

C.1 Corpus and token budget

The corpus ground truth is the public FineWeb (Penedo et al., 2024) `sample-10BT` config on HuggingFace: 95 dated Common Crawl (`CC-MAIN`) dumps spanning `2013-20` through `2024-10`, totalling 1.037×10^{10} tokens. Each document's `dump` field is mapped to the Monday of its ISO year-week to place it on a calendar axis, and the per-dump token totals (cached in `fineweb_10BT_dump_sizes.json`) serve as the denominators for rate normalisation. The sample is unevenly distributed across years—the mid-2010s crawls are the largest and the truncated 2024 slice the smallest—so per-year token totals (Table 3) are essential for interpreting raw counts.

Year	Tokens (M)	Year	Tokens (M)
2013	185.4	2019	1,236.9
2014	714.1	2020	974.6
2015	827.7	2021	1,134.4
2016	765.7	2022	852.1
2017	1,429.7	2023	809.9
2018	1,336.9	2024	103.9

Table 3: FineWeb 10BT sample token totals by calendar year (millions), summed over that year's `CC-MAIN` dumps. Total: 10,371.5 M tokens across 95 dumps. The 2024 slice is truncated at dump `2024-10`.

C.2 Context extraction

For each target word we stream the sample and, for every surviving document, extract a ±20-token window of the original text around each match (`fineweb_context.py`, `-context-window 20`), preserving the raw casing, punctuation, and spacing needed for sense classification. Matching is performed on normalised tokens (whitespace-split, lower-cased, leading/trailing punctuation stripped) via a regex that mirrors the tokeniser's word boundaries, so multi-word targets (*red pill*, *glow-up*) and inflected forms are captured. One CSV of `(target, uri, target_context)` rows is produced per dump.

C.3 Sense disambiguation

Each extracted context is scored with a RoBERTa classifier fine-tuned to separate the *slang* sense from the *standard-English* sense (e.g. *sick* ill vs. excellent, *aura* energy-field vs. Gen-Z coolness). The judged word is injected into a prompt so the decision is conditioned on *which* word is in question: `Is the word "{target}" used as slang here? {context}`. The paper's figures count only occurrences scoring ≥ 0.99; the threshold is a tunable parameter exposed by `roberta_filter.py` (raw scores lie in $[-1, 1]$). Appendix D documents the classifier's data and evaluation.

C.4 Normalisation and uncertainty

Sense-filtered counts are converted to occurrences per million tokens of the corresponding dump, $r_{w,d} = 10^6 \cdot \text{count}_{\text{slang}}(w,d)/\text{tokens}(d)$, using the per-dump token totals above. Each per-dump count is modelled as a Poisson draw $k \sim \text{Poisson}(r_{w,d} \cdot \text{tokens}(d))$, and the figures show the exact (Garwood) Poisson interval (95% by default) as a shaded band; the band tightens as the centred smoothing window aggregates more of the sample.

C.5 Plotting and peak extraction

The band figures (Figures 1–3) use a linear y-axis (the `_count` variant) with a 5-dump centred moving average; the CI band is the exact Poisson interval on the window-pooled count. For the pre-2018 band the y-axis is broken so *lol* does not compress the other curves. Dumps with zero sense-filtered occurrences are dropped so the line connects neighbouring dumps rather than falling to the floor, and a $\log_2$-axis version of each figure is also produced. The corpus peak year (Table 1) is the argmax, over calendar years, of the year-pooled normalised rate (`peak_year.py`). All band figures are generated by `word_rate_plotter.py -highlight-bands`, which pools the target and scenario scored contexts.

C.6 Implementation

`FineWebAnalysis/`: `fineweb_context.py` (context extraction), `roberta_filter.py` (context scoring), `word_rate_plotter.py` (frequency curves and Poisson bands), `peak_year.py` (peak-year extraction).

D Sense Classifier: Data, Training, and Validation

D.1 Model and training

The sense classifier is a `roberta-base` sequence classifier fine-tuned for binary slang-vs-standard judgement (`finetune_roberta.py`). The target word is injected into the prompt template above and the rendered string is truncated/padded to 256 tokens. Training uses a class-weighted cross-entropy loss (weights inversely proportional to class frequency, to offset the label imbalance), AdamW (learning rate 2×10^{-5}, weight decay 0.01), batch size 16, and a linear schedule; we hold out a stratified 20% split (`random_state=42`) and keep the checkpoint with the best validation accuracy. The exact prompt template is serialised next to the model so that `roberta_filter.py` applies identical wording at inference.

D.2 Annotation data

Training data combines human-annotated FineWeb contexts (1,310 rows over 15 words, 649 slang / 661 standard) with a larger synthetically generated set (8,504 rows over 124 words, 6,108 slang / 2,396 standard). Synthetic examples are produced by prompting Claude for 50–100 natural example sentences per target word, each self-labelled slang or standard; words with a genuine everyday sense are balanced ∼50/50, while slang-only words (*skibidi*, *rizz*, ...) are all-slang (`generate_synthetic_annotations.py`). Conditioning the classifier on the target word lets a single model generalise across the full target vocabulary rather than requiring a per-word classifier.

D.3 Validation of the synthetic labels

Because the training set leans on synthetic labels, we audited their reliability against human judgement on a 995-example sample (`validation_metrics.py`), treating the manual annotation as ground truth. Synthetic labels agree with human labels 98.7% of the time (precision 99.9%, recall 98.4%, $F_1 = 99.1\%$; false-positive rate 0.4%, false-negative rate 1.6%; confusion matrix TN = 261, FP = 1, FN = 12, TP = 721), confirming the synthetic labels are a faithful stand-in for human sense judgements.

D.4 Held-out evaluation protocol

The trained classifier is evaluated on a separate, class-balanced held-out set of FineWeb contexts (1,340 rows over 15 words, 670 slang / 670 standard; `fine_web_small_validation_set`) via `auc_eval.py`, which reports ROC AUC together with a confusion matrix, precision/recall/F_1 at the operating threshold, and a per-word breakdown. Passing `-train-dir` additionally restricts the metrics to contexts whose text does not appear in the training annotations, isolating performance on genuinely unseen strings. **TODO:** Report the held-out ROC AUC and threshold-0.99 precision/recall from the final `ft_model_roberta` checkpoint.

E Experiment 2: Direct Year Association

These queries share the system message `"Give the response as a single returned token or word."` Two are free-text point estimates and the remainder are logprob/echo queries over the same nineteen target words and the year range 2013–2025:

1. **GiveYear:** "The slang word '{word}' is most associated with what year? Respond using only the year."
2. **DatePhrase:** "You hear someone saying '{word}' in conversation, which year are they most likely saying this in?"
3. **AssociateSlangLogprobs:** "A standout word in term of popularity increase in {year} was {word}" — an echo+logprob probe (`logprobs=5, echo=true`) that scores the joint completion logprob of each *(year, word)* pairing across 2013–2025, yielding a continuous year $\times$ word association surface rather than a single point estimate.
4. **GiveTopSlang / GiveTopSlangLogprobs:** "What is the most unique and popular slang word associated with the year {year}? Respond using only a single word." — the inverse mapping (year $\rightarrow$ word), collected as free text and, for the open models, with echo+logprobs; reported as supplementary.

GiveYear and DatePhrase are rendered once per target word and sampled five times per word on all four models ($19 \times 5 \times 2 \times 4 = 760$ responses); a four-digit year is parsed from each response. The mean responded year per word, pooled over GiveYear and DatePhrase, is plotted against the corpus peak in Figure 4; the per-model mean absolute errors against the corpus peak are 6.5 (Claude), 5.5 (GPT-4o, inflated by its collapse to "2023"), 10.3 (DeepSeek), and 7.1 (Llama) years.

F Experiment 3: Semantic Slot Querying

All slot prompts are sampled at temperature 1.0. Concept-slot prompts use the system message `"Respond with a single word only. No explanation, no punctuation - just the word."`; target-recovery prompts use `"Respond with only a single word or very short phrase - no explanation, no punctuation."`

Concept-slot mode. Six meaning slots, each with four independently worded phrasings (24 prompts total); the model's single-word answer is recorded without naming any target, surfacing its default vocabulary per slot. The full phrasing set:

1. *cool/impressive:* (1) "What's a single informal word someone might use to describe something they think is very impressive or cool?" (2) "Give me one slang word people use to say something is excellent or impressive." (3) "What's an informal word for something that is really good and worth getting excited about?" (4) "Name a single casual word that means something is exceptionally cool or impressive."
2. *mediocre:* (1) "What's a single informal word for something that is average, unimpressive, or not worth your time?" (2) "Give me one slang word people use to say something is mediocre or disappointing." (3) "What's a casual word that means something failed to meet expectations?" (4) "Name a single informal word that means something is just okay — not good, not terrible."
3. *charisma:* (1) "What's a single informal word for someone's natural ability to effortlessly attract others or come across as charming?" (2) "Give me one slang word that describes the quality of being naturally magnetic, charming, or romantically appealing." (3) "What's a casual word for the thing that makes someone just naturally attractive or charismatic without trying?" (4) "Name a single informal word for the kind of confident, effortless charm that makes people drawn to someone."
4. *agreement:* (1) "What's a single informal word someone uses to express strong enthusiastic agreement or approval?" (2) "Give me one slang word people say to mean 'yes, absolutely, that's totally true.' " (3) "What's a casual word for when you hear something and you completely agree with it?" (4) "Name a single informal word used to affirm what someone just said — like a stronger, more casual 'exactly.' "
5. *suspicious:* (1) "What's a single informal word for when something about a person or situation feels off, untrustworthy, or not right?" (2) "Give me one slang word that means something seems off or like it shouldn't be trusted." (3) "What's a casual word for when you sense that someone is being deceptive or hiding something?" (4) "Name a single informal word for a feeling that something is unsafe or that a person can't be trusted."

6. *excellent (sensory):* (1) “What’s a single informal word someone would use to say a song, food, or piece of art is exceptionally good?” (2) “Give me one slang word people use when something sounds, tastes, or feels really, really good.” (3) “What’s a casual word for when a song or meal is so good it gives you a visceral, satisfying reaction?” (4) “Name a single informal word meaning something is so good it almost overwhelms you in a great way.”

Target-recovery mode. Each of the nineteen target words has three phrasings written to evoke that word specifically. A response counts as a recovery if it matches the target after case-folding, applying a light suffix stemmer and whole-word phrase containment — so *glowup*/*glow-up*, *gaslighting*/*gaslight*, and “take the red pill”/*red pill* all count, while *swagger* is not credited as *swag*. Listing 1 shows two of the nineteen templates.

Year-conditioned variant. Each target-recovery prompt is prefixed with “The year is {year}.” and the year is swept over 2013–2025 (13 values). With four models, nineteen words, three phrasings, thirteen years, and ten samples per cell this yields 29,640 responses (7,410 per model; 30 per word × year cell) — the data behind the year-resolved heatmaps (Figure 6). Aggregate target-word recovery (Figure 5) pools these responses over the stated year.

Listing 1: Two of the nineteen target-recovery prompt templates (Experiment 3). The year-conditioned variant sweeps the year list over 2013–2025; the unconditioned recovery figure pools over all years.

```
{ "id": "exp3_lol",
  "system": "Respond with only a single
      word or very short
             phrase -- no explanation,
                 no punctuation.",
  "user": "The year is {year}. {phrasing
      }",
  "temperature": 1.0,
  "variables": {
    "year": ["2013", "...", "2025"],
    "phrasing": [
      "What's a way, using text, people
          might show that
       something made them laugh?",
      "Give me the short response
          someone adds to a message
       to indicate they're laughing.",
      "What phrase do people use online
          to react to
       something funny?"
    ]
  }
}

{ "id": "exp3_aura",
  "system": "Respond with only a single
      word or very short
             phrase -- no explanation,
                 no punctuation.",
  "user": "The year is {year}. {phrasing
      }",
  "temperature": 1.0,
  "variables": {
    "year": ["2013", "...", "2025"],
    "phrasing": [
      "What's a single informal word for
          the intangible
       coolness or presence someone
          radiates?",
      "Give me one word young people use
          for the invisible
       'points' you gain or lose by
          acting cool or cringe.",
      "What word for an invisible energy
          field around a
       person is now slang for their
          coolness or charisma?"
    ]
  }
}
```

G Models and Sampling

1. `meta-llama/Llama-3.3-70B-Instruct-Turbo` — Together AI; open weights, token logprobs available.
2. `deepseek-ai/DeepSeek-V4-Pro` — Together AI; open weights, token logprobs available.
3. `gpt-4o` — OpenAI; closed, no token logprobs.
4. `claude-sonnet-4-6` — Anthropic; closed, no token logprobs.

A single harness (`PromptingSlang/`) dispatches each prompt to the correct provider by model-id convention (`org/model` → Together, `claude-*` → Anthropic, `gpt-*/o#-*` → OpenAI), retrying with exponential backoff. Prompts expand by the Cartesian product of their `variables` block, and each expansion is sampled `-samples` times (an additional `-closed-samples` factor lets closed models approximate a token distribution by repeated sampling); generation is capped at 512 tokens. All probing prompts set temperature 1.0 per template. Because the closed APIs do not expose token logprobs, their response distributions are approximated by repeated sampling, and the logprob/echo queries run only on the open-weight models. Runs are keyed by `(model, prompt, sample index)` and are resumable, so partial runs top-fill rather than restart. Prompt templates and the response-collection harness are in the `PromptingSlang/` directory of the project repository.

H Naturalistic Generation: Statistical Analyses

This appendix details the persona-simulation probe of Experiment 4 (Section 4.4, Figure 7) and the estimators that quantify its "temporal vintage."

H.1 Probe and vocabulary

Each model is asked to role-play informal peer conversation via two prompts (system: `"You are a young person using informal language like you would among peers. . . "`; e.g. "Simulate a brief conversation between two or more teenaged friends taking place online or over text message"), yielding 100 responses per model (400 total). Generated text is matched against a 150-word slang vocabulary (the union of `target_words.txt` and `scenario_words.txt`), each word carrying the corpus peak year of Experiment 1. As in the forced-choice queries, output concentrates on a handful of high-frequency, older terms: pooled across models the top items are the skull emoji (`U+1F480`), *u*, *lol*, *omg*, and *bro*, while post-2020 slang appears with counts near 1.

H.2 Effective vintage

We summarise each model's produced slang by its *effective vintage*: the count-weighted mean corpus-peak year of the slang it emits (`effective_vintage.py`). Over the vocabulary of 88 words with ≥ 100 corpus hits (31 of which appear across the 400 pooled responses, 1,848 in-lexicon hits), the model vintage is **2014.0** (95% CI $[2013.9, 2014.1]$; parametric Poisson bootstrap, $B = 5000$, seed 42) against a frequency-weighted corpus vintage of **2015.8** — a lag of -1.7 years, i.e. models skew older than the corpus baseline over the same vocabulary. The high-frequency skull emoji (`U+1F480`) is excluded from the vintage estimators, as its corpus rate is unreliable under FineWeb's text-processing.

H.3 Temporal mixture

As a distributional cross-check we fit the model's produced-year distribution as a mixture over calendar-year slices (2013–2024) and read off an effective vintage from the fitted weights (`temporal_mixture.py`, $\varepsilon = 0.5$ smoothing). Both fits concentrate the weight in the mid-2010s: the EM (KL-minimising) fit puts $\approx$0.73 on 2016 with the remainder on 2022 (effective vintage **2016.0**), lowering the divergence from a uniform baseline from 0.82 to 0.63 nats, and the L2 fit places all weight on 2016 (also **2016.0**). Per-word residuals show the model over-indexing on a few mid-era terms (*bro*, *lit*, *bruh*) and under-indexing on genuinely recent slang (*legit*, *vibe*, *vibes*).

H.4 Regression cross-check

A negative-binomial regression of per-word response counts on corpus peak year (`nb_regression.py`, 89 words) confirms severe overdispersion (dispersion $\alpha \approx 11$; likelihood-ratio test vs. Poisson $p < 0.001$), which is why we rely on the vintage and mixture estimators for the headline comparison: with 57 of 89 words never produced and a handful of high-frequency recent outliers (*legit*, *vibe*), the regression's point estimate is unstable and we do not draw a directional claim from it. **TODO:** Report the response-level (rather than parametric) bootstrap variant of the effective-vintage estimate.